PHYS 206 Experiment 2 Lab Report: Young's Double Slit Experiment

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1 Introduction

We will observe and analyze the interference pattern formed by light passing through two closely spaced slits and to determine the wavelength of the laser light used in the experiment.

In 1801, Thomas Young successfully measured the wavelength of light using a double-slit experiment. His experiment was instrumental in supporting the wave theory of light, countering Newton's corpuscular (particle) model. Today, you will attempt to replicate this experiment using monochromatic laser light. When light shines on an opaque screen with two closely spaced, narrow slits, different theories predict different outcomes:

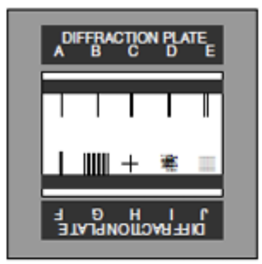
Newton's particle model Light should behave like a stream of particles, producing two bright bands directly behind the two slits.

Huygens' wave model Each slit acts as a new source of light waves. Since the waves originate from the same source, they are coherent and interfere with each other. This results in a pattern of alternating bright and dark fringes on the screen.

2 Equipment

The following equipment was used:

- Optics bench
- Laser source
- Diffraction plate (Figure 1)
- Component holders
- Viewing screen
- Measuring tape and ruler
- Calculator



Pattern	No. Slits	Slit Width (mm)	Slit Spacing center-to-center (mm)
A	1	0.04	
B	1	0.08	
C	1	0.16	
D	2	0.04	0.125
E	2	0.04	0.250
F	2	0.08	0.250
G	10	0.06	0.250
H	2 (crossed)	0.04	
I	225 Random Circular Apertures (.06 mm dia.)		
J	15 x 15 Array of Circular Apertures (.06 mm dia.)		

Figure 1: Diffraction Plate

3 Interference Pattern

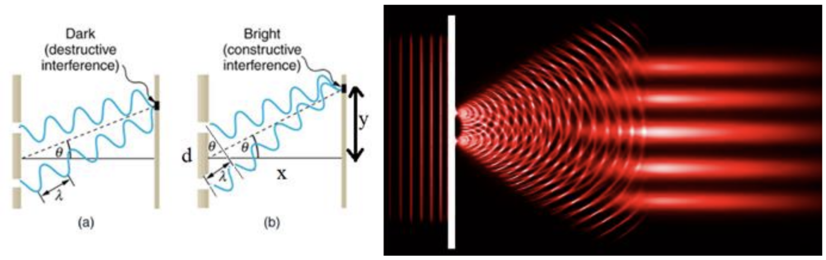


Figure 2: Young's double slit interference pattern

For contructive interference;

$$\sin\theta = \frac{m\lambda}{d} \quad (m = 0, 1, -1, 2, -2...) \tag{1}$$

Where λ is the wavelegth of the light source, where d is the distance between the slits, θ is the angle relative to the incident direction, and m is the order of the interference.

Given that the distance between the first and second maxima is y and the distance from the diffraction plate and the screen is x;

$$\tan\theta = \sin\theta = \frac{y}{x} \tag{2}$$

Since x is quite large with respect to y, then θ becomes a very small angle and we can use the small angle approximation: $\tan\theta \approx \sin\theta \approx \theta$ in the above formulas.

From Eq. 1 and 2;

$$\lambda = \frac{yd}{x} \tag{3}$$

For higher orders of interference;

$$m\lambda = \frac{y_m d}{x} \tag{4}$$

4 Procedure

1. The set up of the optics bench with laser and diffraction plate should be available on the tabletop.
2. After checking the equipment, adjust the plate so that the laser light is passing through one of the double slits and an interference pattern is produced on the screen.
3. Measure the slit spacing, d and record it to the data table.
4. Measure the slit width l and record it on the data table.
5. Measure the distance from the plate to screen, x and record it to the data table.
6. Measure the distance of each bright spot to the the center bright spot, y and record in the Table 1.
7. Repeat the procedure for two more slits and record your measurements on Table 2 and Table 3.

5 Data Analysis

Table 1: Slit D

Slit Spacing center-to-center d (mm)	**Slit Width** l (mm)	**Screen Distance** x (cm)
0.125	0.04	50

Fringe Order m	**Fringe Spacing** y_m (mm)	**Wavelength** λ (nm)
1	2.25	632.8
2	4.5	632.8
3	7	632.8
4	9.2	632.8

Table 2: Slit E

Slit Spacing center-to-center d (mm)	**Slit Width** l (mm)	**Screen Distance** x (cm)
0.250	0.04	79.5

Fringe Order m	**Fringe Spacing** y_m (mm)	**Wavelength** λ (nm)
1	2.1	632.8
2	4.3	632.8
3	6.5	632.8
4	8.6	632.8

Table 3: Slit F

Slit Spacing center-to-center d (mm)	**Slit Width** l (mm)	**Screen Distance** x (cm)
0.250	0.08	78

Fringe Order m	**Fringe Spacing** y_m (mm)	**Wavelength** λ (nm)
1	2.1	632.8
2	4.2	632.8
3	6.4	632.8
4	8.5	632.8

By using Eq. 4, experimental wavelength λ_{exp} can be calculated.

$$\lambda_{exp} = \frac{y_m \cdot d}{m \cdot x}$$

Averaging the calculated wavelengths of each slit are followings:

- **Slit D:** 570.8 nm

- **Slit E:** 673.5 nm
- **Slit F:** 677.8 nm

The overall average is 640.7 nm.

Comparing this to the wavelength of our laser ($\lambda_{th} = 632.8$ nm), the percent error is calculated as:

$$\% \text{ Error} = \left| \frac{\lambda_{th} - \lambda_{exp}}{\lambda_{th}} \right| \times 100$$

$$\% \text{ Error} = \left| \frac{632.8 - 640.7}{632.8} \right| \times 100 \approx 1.25\%$$

6 Discussion

Questions

1. **How does changing the slit separation affect the interference pattern?**

   Answer: Based on the equation $y_m = \frac{m\lambda x}{d}$, y_m is inversely proportional to d and proportional to x. So, decreasing d increases the fringe spacing if x is constant. Even though in the table y_m seems unchanged x was increased so our experimental results for slit D and slit E confirm that.

2. **How does changing the slit width affect the interference pattern?**

   Answer: Changing the slit width does not affect the interference pattern but it changes the diffraction pattern. A narrower slit would cause the overall pattern to spread out more and changes positions of the dark fringes that has to be bright if only interference is considered.

3. **What happens if white light is used instead of laser light?**

   Answer: White light contains a continuous spectrum of visible wavelengths. Because the diffraction angle depends on wavelength by Eq. 1, different colors diffract at different angles. Therefore, fringes will spread out into rainbow spectra, with violet closest to the center and red furthest.

4. **What assumptions are made in the small-angle approximation?**

   Answer: The small-angle approximation assumes that $\sin\theta \approx \tan\theta \approx \theta$. For this to be mathematically valid, the distance from the slits to the screen (x) must be significantly larger than both the slit separation (d) and the fringe spacing on the screen (y). This help us to substitute y/x for $\sin\theta$.

5. **How could errors in measuring affect the final calculation of λ?**

   Answer: Actually, our error is very very small, so it is probably caused by our measurement instrument because we use a kind of regular ruler to measure it and it not precise enough.

6. **How does the double-slit experiment support the wave-particle duality of light?**

   Answer: In our experiment, light behaves like a wave. If it behaved as a particle, we would expect regular shadows but we saw a pattern caused by wave property. Although our setup does not show the particle property of light, we can design a double-slit experiment with two polarizer and put them in front of the slit with different polarization angles, they will behave like particle because of the uncertainty and the interference patter would disappears.

7. **What happens to the interference pattern if we try to measure which slit a photon passes through?**

   Answer: As I mentioned in Question 7, if we measure one of the photon, we cannot see the pattern because the uncertainty principle does not let it behave as a wave since the path the photon passing through is known, now.